

Support-Free Thermally Insensitive Hollow Core Fiber Coil

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Abstract—Light traversing an optical fiber is subject to various local phase perturbations driven by temperature. This thermal phase sensitivity is undesirable in fiber interferometers and their applications which require that a fixed, stable phase be received after propagation. The use of hollow core fiber (HCF) has been shown to reduce this thermal phase sensitivity over solid core fibers and here we propose and demonstrate how coiling HCF to a prescribed geometry can further significantly reduce this sensitivity. Our proof-of-concept experiment shows reduction by a factor of ~ 90 with respect to the uncoiled HCF, and over three orders of magnitude with respect to uncoiled solid core optical fiber. Our strategy exploits a nuance of the elastic properties of fiber coils whereby the constrained thermal expansion of the composite material (fiber + coating) can result in a coil having compressed inner layers and expanded outer layers. Thermal expansion is the dominant effect responsible for thermal phase sensitivity in HCFs, and in this scheme the compressed inner coil layers compensate the thermal expansion of the outer layers. In this study we design the coil parameters using finite element simulations, studying the relationship between coil performance and its key parameters. The proof-of-principle coil has 160 mm diameter and incorporates a 548 m length of HCF out of which a 230 m section shows almost zero (slightly negative) thermal phase sensitivity. Though the coil shows low thermal phase sensitivity over tens of hours, the long-time constant viscoelastic properties of the coating materials used in the HCF under study are shown to limit these benefits. To make this strategy practical for systems with fast temperature dynamics, a coating having more stable mechanical properties could be used. For precision timing systems in which long thermal time constants are already the norm, this scheme represents a low-cost and provides a significant reduction to thermal sensitivity which is immediately practicable.

Index Terms—Hollow core fiber, optical fibers, thermal phase sensitivity.

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I. INTRODUCTION

OPTICAL fibers are widely used in applications such as telecommunication, sensing, and laser beam delivery. When light is transmitted through a fiber, the accumulated optical phase of the light received changes with the fiber temperature. This effect limits the performance of most fiber interferometers and hence correspondingly their applications, e.g., in laser frequency stabilization or fiber sensing, where cross sensitivity to temperature limits the achievable performance [1], [2], [3], [4], [5], [6]. This limitation in key applications highlights the need for strategies to reduce or even completely null this thermal phase sensitivity.

The accumulated phase of the signal propagating through an optical fiber is:

$$\varphi = 2\pi n_{eff}L/\lambda, \quad (1)$$

where L is the fiber length, n_{eff} is the effective refractive index of the fundamental mode, and λ is the light wavelength in vacuum. Consequently, the fiber’s thermal phase sensitivity, normalized to fiber length, can be defined as:

$$S_\varphi = \frac{1}{L} \frac{d\varphi}{dT} = \frac{2\pi}{\lambda} \left(\frac{n_{eff}}{L} \frac{dL}{dT} + \frac{dn_{eff}}{dT} \right). \quad (2)$$

The first term on the left-hand side of (2) describes the effect of fiber elongation due to thermal expansion, while the second term quantifies the effect of refractive index change, which is due mainly to the thermo-optic effect [7], [8].

Various methods have been demonstrated to reduce the thermal phase sensitivity in standard single-mode optical fibers (SMFs) that guide light through a glass core. In SMF, the thermo-optic effect contribution is significantly larger (95%) as compared to the thermal expansion (5%) [8] and thus strategies to reduce SMF thermal phase sensitivity focus on reduction of the thermo-optic coefficient. Such techniques include core doping to directly reduce the material thermo-optic coefficient [9], or application of specialty coatings [10] to introduce a temperature-dependent stress, and thus an elasto-optic refractive index variation which compensates the variation arising from the thermo-optic effect.

An alternative approach is to use hollow core fibers (HCFs), in which the light is mostly transmitted through air. As the thermo-optic coefficient of the air at constant volume is orders

of magnitude smaller than in silica glass, the second term in (2) is negligible in HCF. Various studies report thermal phase sensitivity in HCF as 3–18 times improved over SMF [11], [12], with the efficacy mostly limited by effects of the HCF coating on the overall thermal sensitivity. This effect was minimized in [13] by using HCF with a very thin coating, achieving 30 times improvement. Although this represents a significant improvement, further reduction of the thermal sensitivity is desirable. Published methods include operation of the HCF at low temperatures, e.g., for uncoated silica HCF at -71°C , where the silica coefficient of thermal expansion (CTE) crosses zero [14]. HCF with open ends was also demonstrated to enable zero thermal sensitivity. Heating HCF increases the pressure inside it, causing air to escape from the core, and thereby making the second term of (2) negative. At 113°C and atmospheric pressure, the first and the second terms have the same magnitude, but opposite sign, resulting in zero HCF thermal sensitivity [15]. Both these methods, however, are not applicable at room temperature. To this end, a method operating at room temperature has been proposed recently [16]. It is based on winding the HCF on a spool made of material with near-zero CTE. This could enable zero HCF thermal phase sensitivity at room temperature, and the initial demonstration showed a reduction by a factor of over 3. Although promising, the coil adds weight and volume, which are undesirable in space-constrained or weight-sensitive applications.

Here, we propose and demonstrate a new method to obtain an HCF with near-zero thermal sensitivity. Our technique represents an efficient, compact, cost-effective deployment of the fiber which exploits the fibers intrinsic material properties to achieve a nulled thermal phase sensitivity, and it does not require the use of low-CTE materials. It relies solely on winding the HCF into a free-standing coil – a deployment which is readily achievable using even very simple fiber winding techniques. Compact fiber coils are the main sensing element for fiber optic gyroscopes [17], [18], and thus their thermo-mechanical properties are well-studied. In such a fiber coil, the different mechanical properties of the silica glass and the fiber coating materials [19] form a composite, orthotropic material in which the competing stiffnesses and thermal expansion coefficients can result in different regimes of overall behavior. Crucially, for certain coil geometries and coating material properties, a temperature increase can result in the innermost coil layers being driven into compression while the outer layers are in tension. We show that this layer-by-layer distribution of circumferential strains can be well-modeled and used to create an HCF path over which the aggregate phase becomes temperature insensitive as the negative strain of the inner layers compensates the positive strain of the outer layers. HCF is ideally suited for this purpose because the mode has minimal overlap with the glass and thus experiences negligible stress-optic effect. Thus, the stresses in the coil layers have minimal effect, and the phase sensitivity can be effectively nulled. In this work we first develop a finite element model of the coil and show how various coiling parameters influence the coil's thermal sensitivity. Subsequently, we build such coil and perform a proof-of-principle experiment which demonstrates the reduced thermal phase sensitivity.

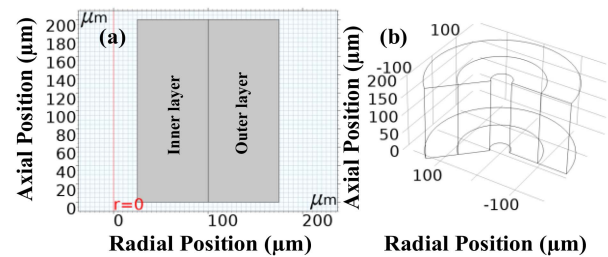


Fig. 1. 2D axisymmetric model (a) Used to simulate a 3D cylinder shell (b). It is made of an Inner layer (young's modulus = 0.55 MPa, CTE = 8 ppm/ $^\circ\text{C}$, thickness: 75 μm); and an outer layer (young's modulus = 73100 MPa, CTE = 0.5 ppm/ $^\circ\text{C}$, thickness: 75 μm).

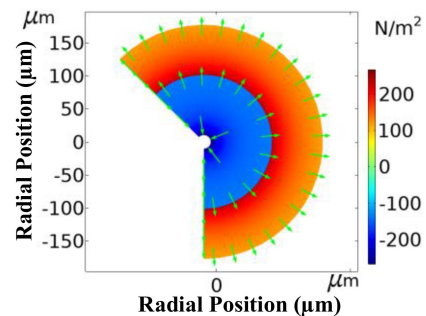


Fig. 2. Deformation and stress distribution of the two-layer cylinder shell (Fig. 1) cross section after the temperature has been increased from 30°C to 40°C . Here, the arrows show the deformation direction. The inner side of the inner layer shrinks, introducing a negative inner diameter change of -14.7 nm (negative stress), while the outer boundaries of the inner layer and outer layer both expand (by $+1.2\text{ nm}$ and $+2.0\text{ nm}$, respectively, positive stress).

II. PRINCIPLE OF OPERATION

To illustrate the principle of our method, we simulate the thermal expansion properties of a simple toy model consisting of a cylindrical shell of two layers made of two different materials. This 3D cylinder shell is modelled using the 2D axisymmetric model in COMSOL Multiphysics. The model and its parameters are given in Fig. 1. Boundary conditions on the top and bottom of this structure are set to zero expansion in the axial direction, which allows us to simulate an infinite cylinder shell.

The initial temperature is set to 30°C and then it is increased to 40°C . Fig. 2 shows the deformation (arrows show the direction of the deformation) and stress distribution (color-coded) at the two-layer shell cross-section. While the Outer layer expands (in both, its inner and outer diameter), the situation is different for the Inner layer. Its outer diameter expands by 2.0 nm , but its inner diameter actually shrinks by 14.7 nm .

One can understand this phenomenon intuitively by considering the expansion of each layer: the low CTE outer layer effectively confines the high CTE inner layer, and thus the inner layer is forced to expand in the only direction available – inward, as confirmed by the stress distribution shown in Fig. 2.

III. SIMULATIONS

Our coil structure is more complex than the two-layer cylinder shell described in the previous section, however, as we show

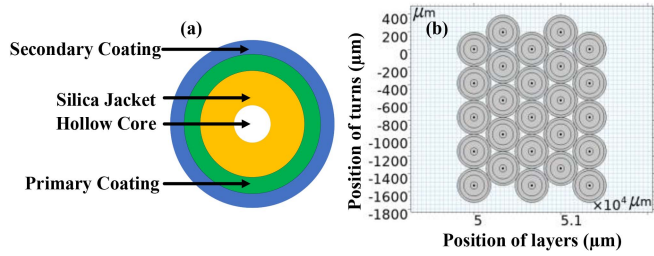


Fig. 3. (a) Simplified HCF structure used in our models (materials shown in Table I) (b) Cross-section of a coil (100 mm diameter) with five layers and five turns.

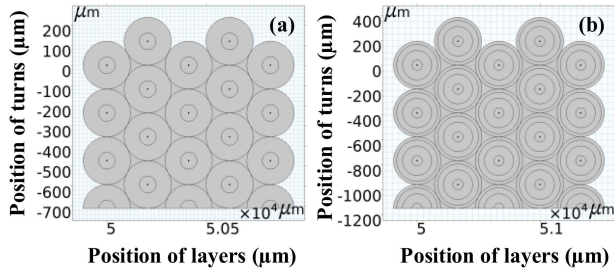


Fig. 4. HCF semi-coil model with (a) Bare HCF and (b) Dual-coated HCF. (Coil diameter: 100 mm; number of layers: 5).

later, it also achieves a zero-expansion point through interplay of materials with different thermal properties.

Our HCF consists of four materials (Fig. 3(a)): air, silica glass, primary coating, and secondary coating. We neglect the influence of the HCF microstructure used for light guiding, as it contains a minimal amount of glass and thus its thermal and mechanical properties have negligible influence on the overall thermo-mechanical HCF properties. The microstructure also has a negligible influence on the optical properties since the fraction of light that propagates through the microstructure in the HCFs we use in our experiments is very small ($<0.01\%$ [20]). This simplifies the HCF structure to four-layers as shown in Fig. 3(a).

Previous work [19] showed that such a coil, with its multiple materials, can behave similarly to our cylinder shell model. Here the strands of non-compliant, low-CTE silica distributed through the composite coil structure constrain the high CTE polymer coating, effectively forcing the innermost layers further inward and into compressive circumferential strain.

The coiled HCF model is shown in Fig. 3(b). We adopt a 2D-axisymmetric model, effectively simulating each fiber turn as a solid torus of glass enclosed by hollow toruses of coating material. While in practice, the fiber coil is a layered helix, this torus model gives a faithful representation of the coil behavior while providing a significant reduction in the required simulation size. The example shown in Fig. 3(b) is of a fiber coil with five layers and 5 turns wound with a 100-mm inner diameter. The used materials are shown in Table I.

To further reduce computing time, we reduce the model size to a semi-coil as shown in Fig. 4, taking advantage of symmetry about the coil mid-plane. We consider coils made of an uncoated (Fig. 4(a)) and coated (Fig. 4(b)) HCF, to further highlight the difference in stress experienced by composite versus mono-material structures.

TABLE I
TYPICAL MATERIAL PROPERTIES OF OPTICAL FIBER AND THEIR COATINGS, AS CONSIDERED IN OUR SIMULATIONS

PARAMETERS	SILICA JACKET	PRIMARY COATING	SECONDARY COATING
Young's modulus (MPa)	73100	0.55	700 (20°C) 520 (40°C) 470 (60°C)
CTE (ppm/°C)	0.47	363	80
Poisson's ratio	0.17	0.37	0.37
Density (kg/m ³)	2203	1030	1130

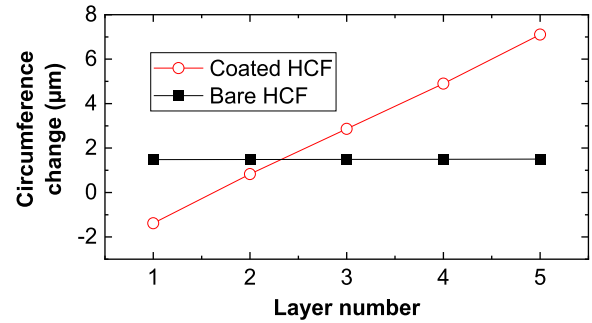


Fig. 5. Average circumference change of each layer in an HCF coil calculated using the model shown in Fig. 4 (bare HCF, black dots; dual-coated HCF, red circles) subject to a temperature change from 30 °C to 40 °C.

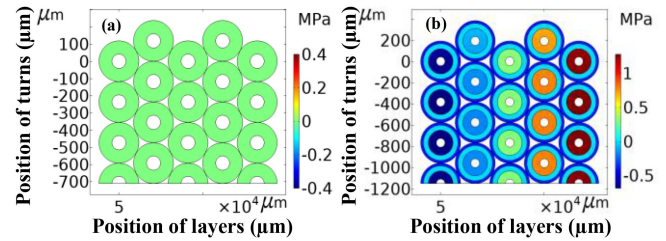


Fig. 6. Stress distribution of the model in Fig. 4(a) Bare HCF (b) Dual-coated HCF.

In the simulation, we change temperature from 30 °C to 40 °C, record changes of the diameters of the air cores at each layer, and calculate the average change in the circumference of each layer, as shown in Fig. 5. When the coil is made of air and silica glass only (bare HCF), all coil layers experience similar HCF elongation, as we would expect. However, the behavior is different for the coated HCF and is akin to the two-material cylinder model. Here, the first layer shrinks, the second one experiences almost no length change, while the outer three layers expand. This difference in their performance can be further confirmed by the stress distribution in Fig. 6, where the bare HCF model only shows zero stress while the coated HCF model sees a very complicated stress distribution: negative stress in layer 1, close to zero stress in layer 2 and positive stress in layer 3, 4 and 5.

To obtain a fiber path with minimal net thermal sensitivity, we use the N inner layers of the coil that, when concatenated, have near-zero change to the total length. This is demonstrated in Fig. 7, which shows the fractional change in path length for concatenated layers (starting from the innermost layer 1). This is

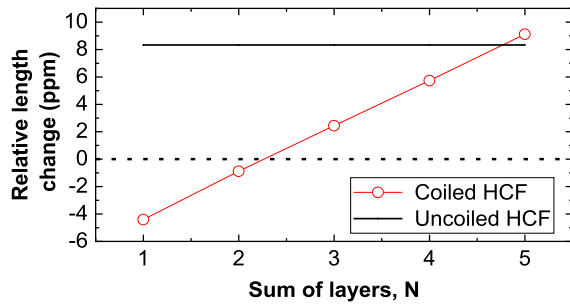


Fig. 7. Calculated dual-coated HCF relative length change when temperature is increased from 30 °C to 40 °C. For coiled HCF (red circles), this change is shown for the inner N layers of the 5-layer coil, Fig. 4(b). For comparison, the change of uncoiled HCF is shown as a black solid line.

compared with the uncoiled HCF behavior, in which the thermal expansion is unconstrained and thus constant.

For the HCF coil modeled here, the sum of layers 1 and 2 produces a near-zero length change over temperature. To build a fiber path having minimal thermal sensitivity we use only the layers giving the minimum thermal expansion for light propagation (layer 1 + layer 2 in our example above), while the rest of the layers (3, 4, and 5 in the example above) are not used for light guiding and serve only to keep the inner layers at minimum thermal expansion.

Although the above example would use only two of five layers, other coil geometries and combinations of material properties yield structures in which the compressive and tensile strains are more balanced about the radial midpoint, thereby creating a minimum thermal expansion path which is closer to the entire coil length. Because we aim to use as much HCF path as possible, this is our figure of merit, R , which we refer to as “coil performance”. In the example above, we used two out of five layers for light guiding, representing $R = 2/5 = 0.4$. In the following sections we investigate how to maximize R , by simulating its dependence on turn count, layer count, coil diameter, and the gap between fibers in consecutive turns.

A. Number of Turns

In general, the coil is unconstrained at the top and bottom surfaces. Thus, the key mechanical constraint is solely in the radial direction, and one expects that within a given layer, the axial position of the turn should have little influence on its circumferential strain. This assumption is supported by the calculated results shown in Fig. 8 where the model shown in Fig. 4(b) was extended to up to 32 turns. Thus, in the following analysis we use only the 4 turns model (as shown in Fig. 4(b)) that is relatively fast-to-run. In the following sections we use it to study the coil performance dependence on coil diameter, number of layers, and fiber gap.

B. Coil Diameter and Number of Layers

The diameter and layer count of the coil is constrained by the volume requirements of the application (for large coils), and the bend-loss performance limits (for small diameters).

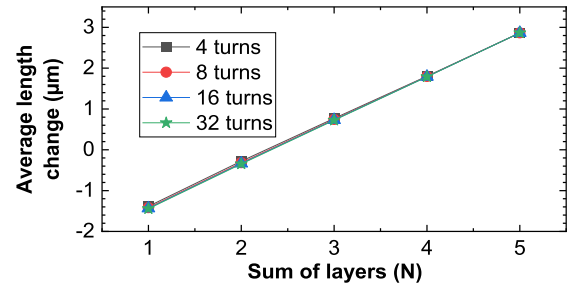


Fig. 8. Average length change in layers 1 to N (sum of layers) for different number of turns (coil diameter: 100 mm; number of layers: 5) subject to a temperature change from 30 °C to 40 °C.

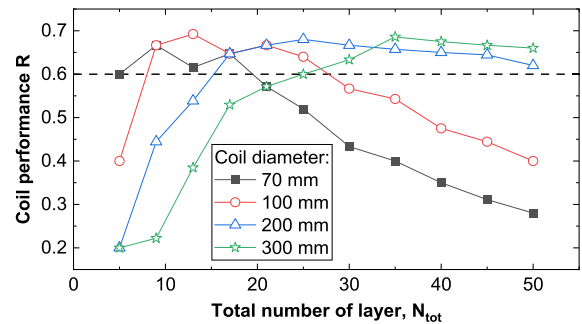


Fig. 9. Simulated coil performance R for different total number of layers N_{tot} and coil diameters of 70 mm (black squares), 100 mm (red circles), 200 mm (blue triangles), and 300 mm (green stars) subject to temperature change from 30 °C to 40 °C.

To keep our analysis within practical limits, we consider coils with diameters from 7–30 cm. Fig. 9. shows the coil performance, R , for different layer counts, N_{tot} , and for a range of coil diameters. This can guide us in choosing the optimum number of layers once an application coil diameter has been determined. For example, for a coil diameter of 100 mm, N_{tot} should be around 13, which achieves a coil performance R of 0.69. The maximum achievable R slightly changes with coil diameter; however, this difference is only moderate (e.g., $R_{\text{max}} = 0.67$ for 7-cm coil diameter versus 0.69 for the 300-mm coil).

C. Fiber Gap

When an optical fiber is wound, a gap is introduced between adjacent turns to enable a smooth and regular winding pattern. In Fig. 10, we model this gap and show how it influences coil performance R . Here we see that the zero gap gives the best performance, while increasing the gap diminishes R down to a value of about 0.4. Performance improves for larger gaps and it should reach the no-gap value when the gap is as large as the fiber diameter (in which case the second layer would just fill the gaps in layer 1, essentially making a single layer with no gap). We expect the performance could improve when a filler (glue) with optimized properties (Young’s modulus and CTE) is applied in between the fibers. However, we do not study this here.

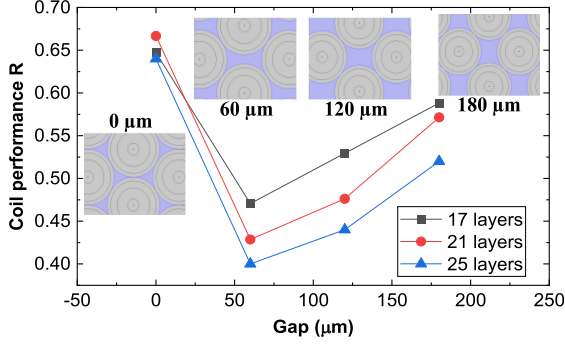


Fig. 10. Simulated influence of the gap distance on coil performance R (coil diameter: 100 mm) subject to temperature change from 30 °C to 40 °C.

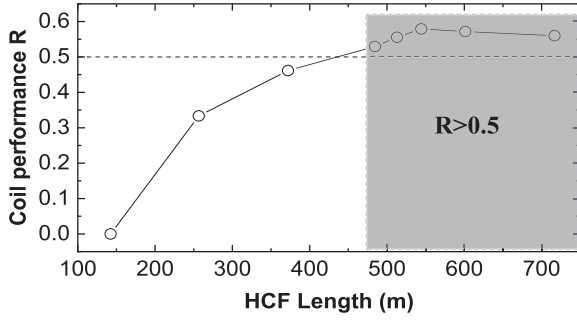


Fig. 11. Simulated influence of the HCF length on coil performance R considering coil realized experimentally (coil diameter: 160 mm; gap distance: 116 μm , turns: 57).

IV. EXPERIMENTS

A. Coil-Winding and Its Parameters

The HCF used was a dual-coated Nested Antiresonant Nodeless Fiber (NANF) [20] hollow core fiber. To select an optimum subset of the HCF path for experiments, we performed simulations using the as-coiled parameters mentioned below.

To prepare a ‘support-free’ HCF coil, the fiber was wound onto a 160 mm diameter spool, which was then removed from the center of the coil to leave a freestanding structure, bound in places with tape/string to keep the outer layers from popping out. For the coil winding, the gap was set at 116 μm . There was on average 57 turns in a single layer.

In Fig. 11, we show the achievable coil performance R as a function of the HCF length. We observe R to increase quickly with HCF length up until a length of 485 m, where it reaches a value of $R = 0.53$. Beyond this, R only increases to 0.58 at a length of 544 m before slightly decreasing, reaching $R = 0.56$ for an HCF length of 717 m. In our experiment, we used an HCF that was 548-m long, which should, based on our simulations, give maximum R . Given the above-discussed spool diameter, number of turns per layer, and gap, this length of fiber can be spooled in 19 layers.

In Fig. 12, we show the calculated (simulated) thermal sensitivity of $N = 8$ (layers 1-8, and corresponding $R = 0.42$), $N = 10$ ($R = 0.53$), $N = 11$ ($R = 0.58$), $N = 12$ ($R = 0.63$), and $N = N_{\text{tot}} = 19$ (the whole coil). For comparison, we

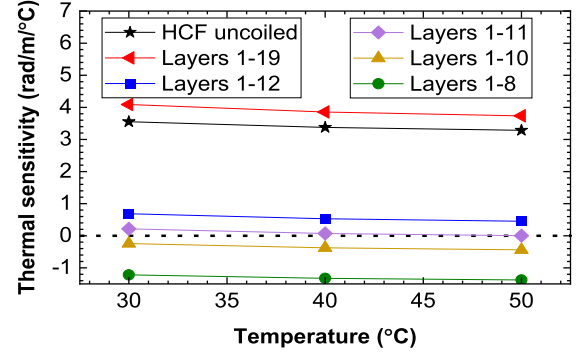


Fig. 12. Simulated thermal phase sensitivity of the experimentally realized HCF coil and its comparison to the non-coiled HCF.



Fig. 13. Photograph of the prepared support-free HCF coil with diameter of 160 mm, pitch of 500 μm , $N_{\text{tot}} = 19$, and 57 turns.

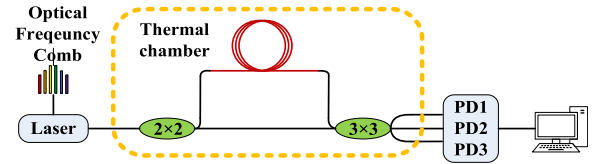


Fig. 14. Experimental setup for thermal sensitivity characterization of the HCF coil (PD: Photodetector, laser was locked to a carrier-envelope-stabilized optical frequency comb (menlo FC1500-250)).

also show the expected result of un-coiled HCF. The minimum thermal sensitivity (sensitivity closest to zero) across the entire considered temperature range is achieved for $N = 11$ ($R = 0.58$).

To validate our predictions experimentally, the coil was wound with access loops at layers 8 and 12. As the optimum occurs at $N = 11$, the model predicted thermal sensitivity of layers 1-8 is negative, while that of layers 1-12 is positive.

A photograph of the prepared coil is shown in Fig. 13. It holds together thanks to the slightly ‘sticky’ nature of the acrylate fiber coating. The Kapton tape was used only to ensure the spool does not ‘break down’ when manipulated and we neglect its influence. To access the fiber of different sections (layer 8 and 12), one-meter of HCF is taken out after spooling layers 8 and then after finishing layer 12 during the spooling process.

B. Thermal Sensitivity Measurements

We measured the thermal phase change of the HCF coil using the Mach-Zehnder interferometer depicted schematically in Fig. 14. The laser (RIO Orion from Luna Technologies)

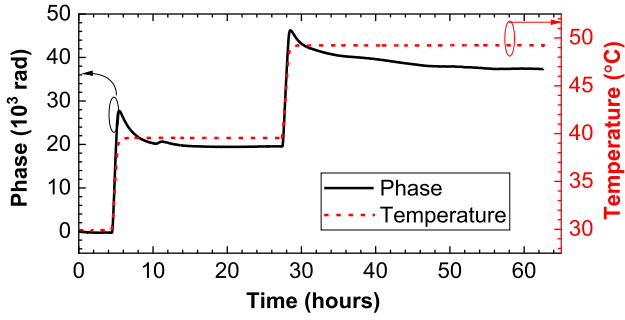


Fig. 15. The measured accumulated phase change of the entire HCF coil (layers 1–19) when the temperature was increased from 30 °C to 40 °C and from 40 °C to 50 °C.

wavelength was locked to a carrier-envelope-stabilized optical frequency comb to avoid any drift and associated interference pattern changes [14]. The entry to the MZI is via a 2×2 fiber coupler and the measurement arm consists of the HCF coil spliced to the coupler pigtailed while the reference arm has the coupler pigtailed directly spliced together. The coupler pigtail lengths were matched in both interferometer arms, so that their contribution to thermally-induced phase delays would cancel in the detected signal. The signals from both arms were combined at a 3×3 coupler and the interference pattern detected by three photodetectors. The use of the 3×3 coupler enables evaluation of the phase, including its sign, from the detected interference fringes, as described in [21]. The entire interferometer was put into a thermal chamber with a thermometer monitoring the temperature inside.

C. Results and Discussion

Firstly, we acquired the thermal phase response of the whole coil (layers 1–19) over two thermal excursions from 30 °C to 40 °C and then from 40 °C to 50 °C. As seen in Fig. 15, the phase initially increases with the temperature. However, once the temperature stabilized, the phase response steadily decreased until it stabilized after some 15–20 hours. This phenomenon of long time-constant phase relaxation arises from the coatings' viscoelastic properties, which were studied in Ref. [13].

After this measurement, we cut at layer 12 and repeated this measurement for the aggregate path spanning layers 1–12. We then did the same for layers 1 to 8. To compare these results obtained for the three different fiber lengths, we use a length-normalized accumulated phase $\varphi_L(t)$:

$$\varphi_L(t) = \varphi(t) / L, \quad (3)$$

where $\varphi(t)$ is the demodulated accumulated phase and L is the HCF length through which the light propagates. Results are shown in Fig. 16 (temperature increase from 30 °C to 40 °C) and Fig. 17 (temperature increase from 40 °C to 50 °C).

To compare the results, we define the thermal phase sensitivity as:

$$S_\varphi = \frac{1}{L} \frac{\Delta\varphi(t \rightarrow \infty)}{\Delta T_t} = \frac{\varphi_L(t \rightarrow \infty)}{\Delta T_t}. \quad (4)$$

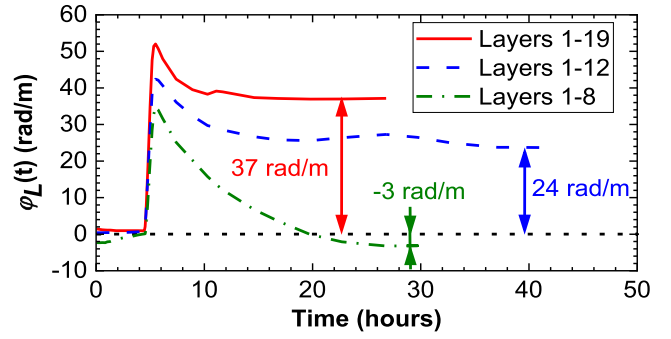


Fig. 16. The measured length-normalized accumulated phase change for $N = 8, 12$, and 19 when subject to a temperature increase from 30 °C to 40 °C.

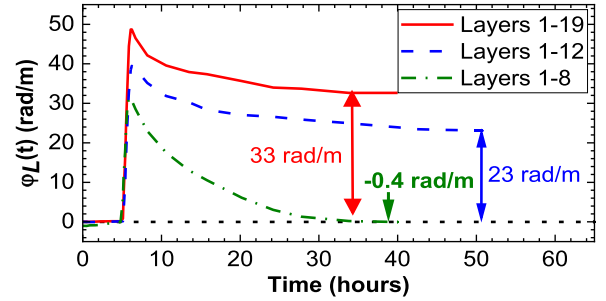


Fig. 17. The measured length-normalized accumulated phase change for $N = 8, 12$, and 19 when subject to temperature increase from 40 °C to 50 °C.

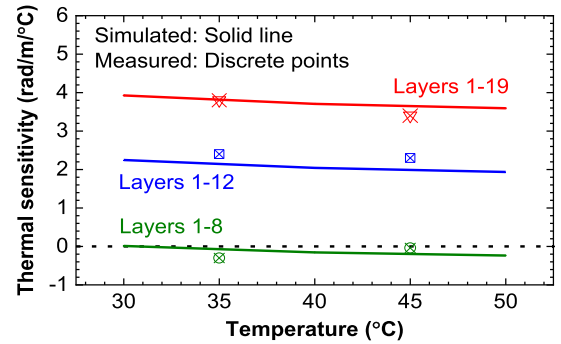


Fig. 18. The measured thermal phase sensitivity of the HCF coil with the simulated results that consider layers 16–19 to be loose.

where ΔT_t is a temperature change applied at $t = 0$. Values derived from measurements are shown in Fig. 18 as discrete points. We see that the thermal phase sensitivity reduces as the layer number N decreases from 19 to 12 and 8. For example, between 30 °C and 40 °C, the thermal phase sensitivity decreases from 3.8 ($N = 19$) to 2.4 ($N = 12$) down to -0.3 rad/m/°C ($N = 8$). This corresponds to the achieved coil performance $R = 0.42$.

These results confirm that the fiber of a few inner layers has a negative thermal expansion coefficient. We also see that the thermal phase sensitivity decreases with temperature, reaching a value of -0.04 rad/m/°C from 40 to 50 °C for $N = 8$. We expect we could tune this closer to zero or even achieve zero thermal sensitivity by unwinding (part of) the outer layers. We also

expect we could reach the zero sensitivity at lower temperatures (e.g., room temperature) by further unwinding the other layers.

The experimental results for layer 1–8 and layer 1–12 (Fig. 18) show less improvement than expected from the simulations (Fig. 12). The desired close-to-zero thermal sensitivity predicted to occur for layers 1–11 was obtained experimentally for layers 1–8. Despite this slight discrepancy, it is important that the desired close-to-zero thermal phase sensitivity has been achieved experimentally.

We found that a good agreement between the simulations and experiment can be obtained when the top four layers from the coil (layers 16–19) are made loose in our model, i.e., they do not contribute to the coil performance. We believe this is not unlikely because the top layers may have become loose in our coil despite securing the spool with Kapton tape, as we have not used any glue and relied only on the fiber sticking together. Another possible explanation is that the coating parameters we used are slightly different to those of the actual fiber.

V. CONCLUSION

With the weak thermo-optic and stress-optic effects, HCF presents a compelling platform for the strong reduction of thermal phase sensitivity in long path interferometers. However, such fibers are still subject to thermal expansion, and thus thermally-induced phase excursions. With the model and experiments we present here, we outline a novel technique to suppress this error by judicious choice of fiber coiling parameters. Because the long fibers used in timing references and other interferometers are already deployed in a coiled form, this technique represents a low-cost intervention which can be deployed merely by prudent choice of fiber coiling geometry. Our coil model confirmed that subsets of a coiled coated HCF can show a zero or even negative thermal expansion, caused by the fiber in the inner coil layers shrinking and fiber at the outer layers expanding as temperature is increased.

Our proof of principle demonstration shows that for relatively typical coil dimensions, one can attain near-zero thermal phase sensitivity. In a 548 m coil of 160 mm diameter, containing HCF wound in 19 layers, 57 turns, and with a fiber gap distance of 116 μm we achieved nearly zero ($-0.04 \text{ rad/m/}^\circ\text{C}$) thermal sensitivity over a temperature step from 40°C to 50°C . This minimum sensitivity was obtained by using only the 230 m length of the inner 8 layers. The observed thermal phase sensitivity is nearly 85 times smaller than that obtained for an uncoiled HCF ($3.4 \text{ rad/m/}^\circ\text{C}$). Considering the thermal phase sensitivity of HCFs is already 14 times smaller than that of standard silica glass core single mode fiber (SMF, [11]), this makes our HCF coil over 1000 times less thermally sensitive than a coil made of SMF. In principle, unwinding some of the outer layers should enable us to reach an even lower value.

We have achieved good agreement between the simulations and experiment by assuming the HCF in the top four layers has loosened, which is a reasonable assumption given the coil is only held together thanks to the slightly sticky nature of the acrylate coating rather than via a glue (which is often used, e.g., in fiber

optic gyroscopes coils). Coil performance may also be improved by optimizing coating properties such as the Young's modulus and CTE.

The relaxation effect due to the viscoelastic behavior of the acrylate coating [13] we observed experimentally (causing temporally phase shift of as much as 33 rad/m, which is almost 8000 rad in the used 230-m long HCF) can be, however, problematic in applications. We expect this to be significantly improved when using a thinly-coated HCF [13] in combination with a glue that would have suitable mechanical properties (low viscoelasticity in particular). Moreover, for applications in thermally isolated environments, where thermal time constants are 15–20 hours or longer, this technique can be employed immediately and at low cost.

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